

Indian Institute Of Technology Gandhinagar



Assignment Report

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1 Task 1: Designing Folded Cascode OTA with NMOS Load

Design a Folded Cascode Operational Amplifier with a single-ended output for the given specifications. Use the current reference to bias the transistors wherever required. Run the simulations for all process corners (tt, ss, ff, sf, fs) and report the obtained specifications for all corners in tabular format.

- Differential Gain > 80 dB
- CMRR > 100 dB
- ICMR: $0.8 - 1.8$ V
- Max. Output Swing: 1 V
- Power $< 20 \mu\text{W}$
- GBW > 10 MHz

The drain current of a PMOS transistor operating in saturation is given by

$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} V_{ov}^2$$

Given:

$$I_D = 4.25 \mu\text{A}, \quad \mu_p C_{ox} = 65 \mu\text{A}/\text{V}^2, \quad V_{ov} = 0.18 \text{ V}$$

The aspect ratio is calculated as

$$\frac{W}{L} = \frac{2I_D}{\mu_p C_{ox} V_{ov}^2} = \frac{2 \times 4.25}{65 \times (0.18)^2} \approx 4.04$$

Therefore, the aspect ratios of transistors M9 and M10 are chosen as

$$\left(\frac{W}{L}\right)_{M9} = \left(\frac{W}{L}\right)_{M10} \approx 4$$

For the NMOS tail transistor operating in saturation, the drain current is given by

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2$$

For the NMOS tail transistor operating in saturation, the drain current is given by

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2$$

Given:

$$I_D = 8.5 \mu\text{A}, \quad \mu_n C_{ox} = 300 \mu\text{A}/\text{V}^2, \quad V_{ov} = 0.2 \text{ V}$$

The aspect ratio is calculated as

$$\frac{W}{L} = \frac{2I_D}{\mu_n C_{ox} V_{ov}^2} = \frac{2 \times 8.5}{300 \times (0.2)^2} \approx 1.42$$

Therefore, the aspect ratio of the tail transistor is chosen as

$$\left(\frac{W}{L}\right)_{\text{tail}} \approx 1.4$$

For the NMOS load transistor operating in saturation, the drain current is given by

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2$$

Given:

$$I_D = 1.5 \mu\text{A}, \quad \mu_n C_{ox} = 300 \mu\text{A}/\text{V}^2, \quad V_{ov} = 0.18 \text{ V}$$

The aspect ratio is calculated as

$$\frac{W}{L} = \frac{2I_D}{\mu_n C_{ox} V_{ov}^2} = \frac{2 \times 1.5}{300 \times (0.18)^2} \approx 0.31$$

Therefore, the aspect ratio of the NMOS load transistor is chosen as

$$\left(\frac{W}{L}\right)_{\text{load}} \approx 0.3$$

For the PMOS load transistor operating in saturation, the drain current is given by

$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} V_{ov}^2$$

Given:

$$I_D = 1.5 \mu\text{A}, \quad \mu_p C_{ox} = 65 \mu\text{A}/\text{V}^2, \quad V_{ov} = 0.18 \text{ V}$$

The aspect ratio is calculated as

$$\frac{W}{L} = \frac{2I_D}{\mu_p C_{ox} V_{ov}^2} = \frac{2 \times 1.5}{65 \times (0.18)^2} \approx 1.43$$

Therefore, the aspect ratio of the PMOS load transistor is chosen as

$$\left(\frac{W}{L}\right)_{\text{PMOS load}} \approx 1.4$$

The biasing voltages used in the folded cascode OTA are:

$$V_{BP} = 1\text{ V}, \quad V_{B4} = 0.83\text{ V}, \quad V_{B6} = 0.72\text{ V}$$

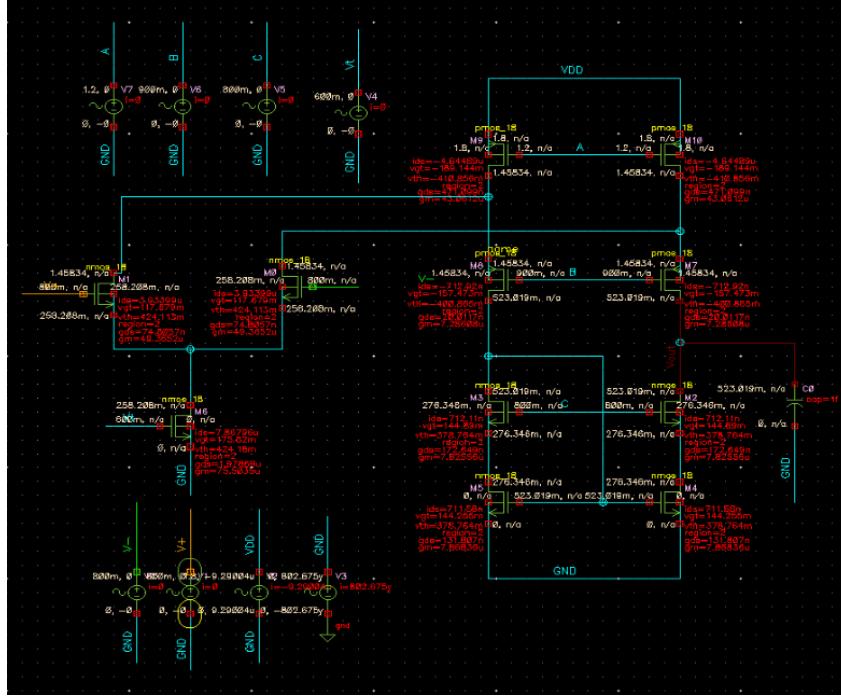


Figure 1: Circuit Diagram of foiled Cascode with NMOS load

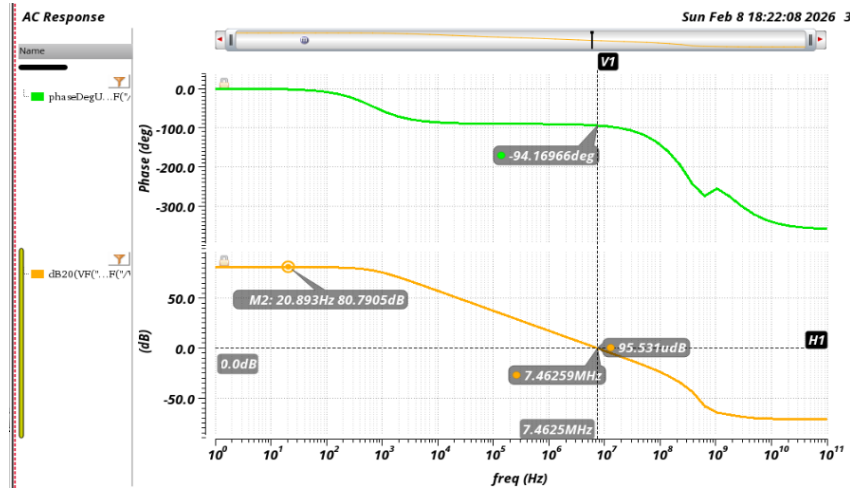


Figure 2: Bode plot for foiled Cascode with NMOS load

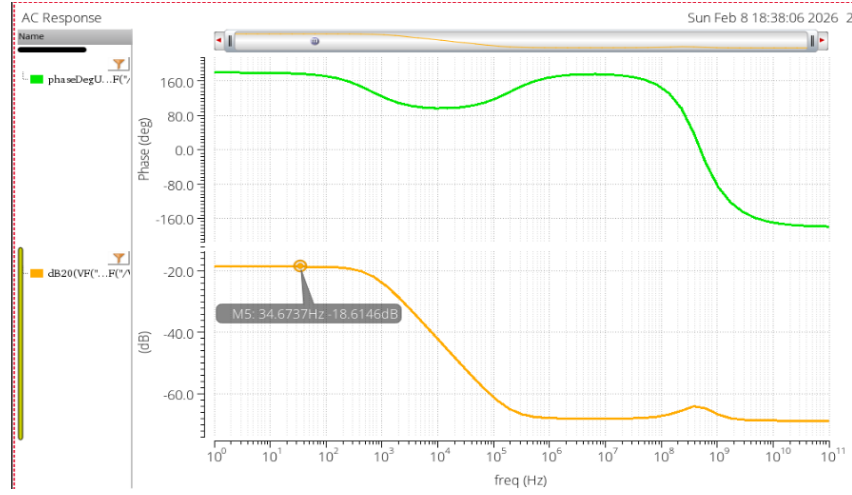


Figure 3: Bode plot for foiled Cascode with NMOS load common mode

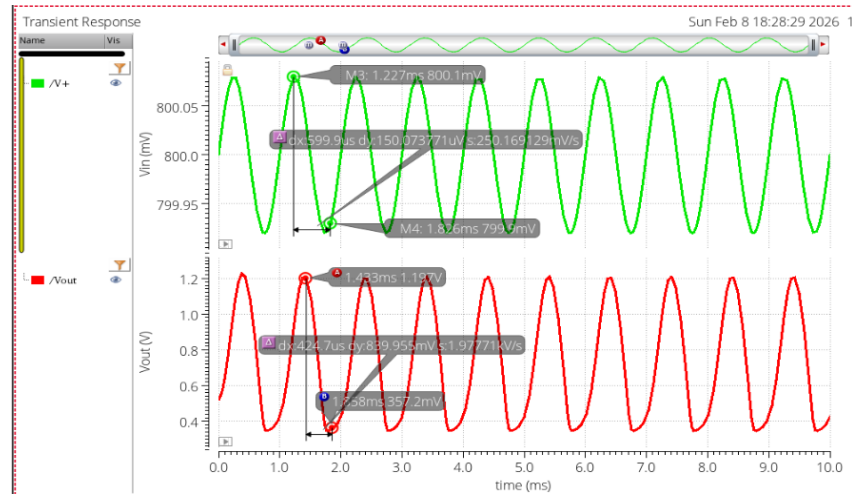


Figure 4: Swing for foiled Cascode with NMOS load

Process Corner	Gain(dB)	CM Gain(dB)	UGB(MHz)	CMRR	Swing(mV)
TT	80.79	-18.614	7.4625	99.404	839.955
FS	-	-	-	-	-
SF	54.458	-27.638	6.204	82.138	736
SS	83.81	-19.72	6.0015	103.51	681.569
FF	58.37	-7.456	15.921	65.826	503

Table 1: Parameters at various Process Corners

1.1 Observations

- The typical–typical (TT) corner exhibits a high DC gain of 80.79 dB with a CMRR close to 100 dB, indicating near-nominal performance.

- The slow–slow (SS) corner provides the highest gain of 83.81 dB and the maximum output swing of approximately 681 mV. However, the unity gain bandwidth reduces to 6.0015 MHz due to lower carrier mobility.
- The fast–fast (FF) corner achieves the highest unity gain bandwidth of 15.921 MHz as a result of faster device operation, but at the cost of reduced gain (58.37 dB) and lower output swing.
- The slow–fast (SF) corner shows degraded performance with low gain (54.458 dB), reduced CMRR (82.138 dB), and lower bandwidth, indicating increased device mismatch effects.
- The fast–slow (FS) corner failed to produce valid results, likely due to biasing or headroom limitations causing improper operation of the folded cascode structure and transistors are out of saturation.
- Power Consumption = $17.2\ \mu\text{W}$

2 Task 2: Designing Folded Cascode OTA with PMOS Load

Design a Folded Cascode Operational Amplifier with a single-ended output for the given specifications. Use the current reference to bias the transistors wherever required. Run the simulations for all process corners (tt, ss, ff, sf, fs) and report the obtained specifications for all corners in tabular format.

- Differential Gain $> 80\ \text{dB}$
- CMRR $> 100\ \text{dB}$
- ICMR: $0 - 1.2\ \text{V}$
- Max. Output Swing: $1\ \text{V}$
- Power $< 20\ \mu\text{W}$
- GBW $> 10\ \text{MHz}$

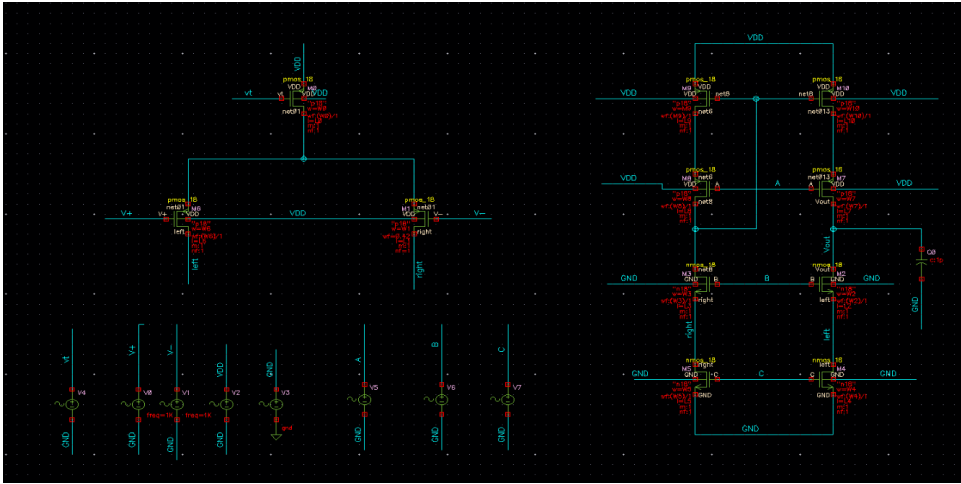


Figure 5: Circuit DIagram for foleded Cascode with PMOS load

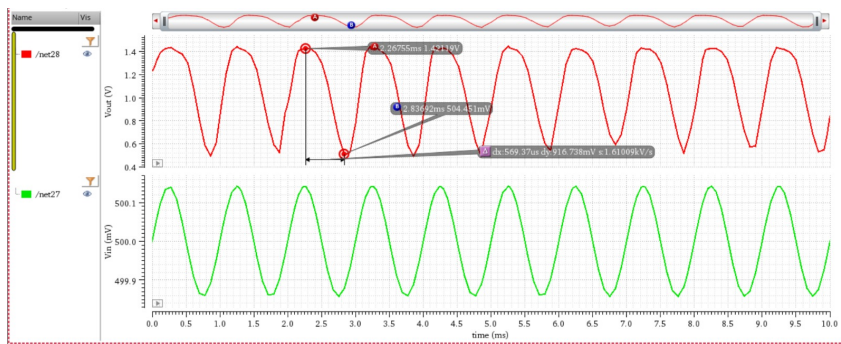


Figure 6: Swing for foleded Cascode with PMOS load

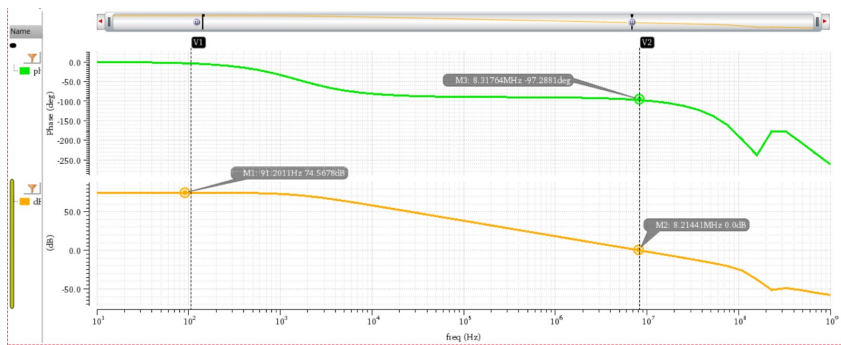


Figure 7: Bode plot for foleded Cascode with PMOS load

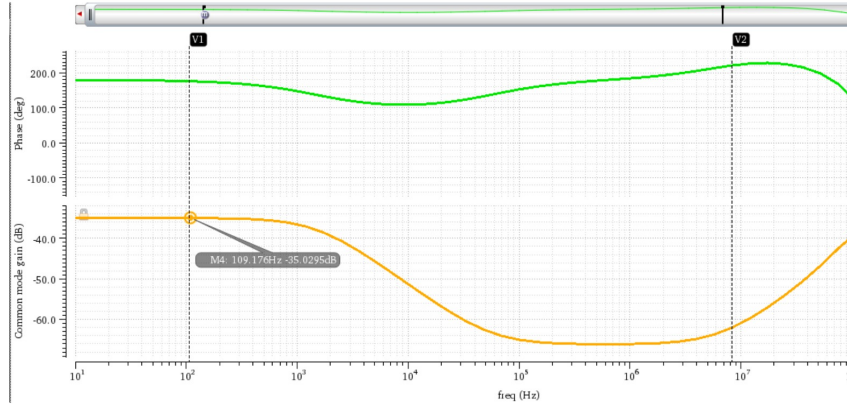


Figure 8: Common Mode gain for foleded Cascode with NMOS load

Process Corner	Gain(dB)	CM Gain(dB)	CMRR	Swing(mV)
TT	74.567	-35.029	109.596	916.8
FS	-	-	-	-
SF	63.2	-27.32	90.52	738
SS	76	-38.2	14.2	890.6
FF	58.37	-29.8	88.17	782.3

Table 2: Parameters at various Process Corners

2.1 Observations

- The typical–typical (TT) corner achieves a gain of 74.567 dB with a high CMRR of 109.596 dB, indicating good common-mode rejection and stable nominal operation.
- The slow–slow (SS) corner shows an increase in gain (76 dB) and output swing (approximately 890.6 mV); however, the CMRR drops significantly to 14.2 dB, indicating poor common-mode rejection under slow device conditions.
- The fast–fast (FF) corner exhibits reduced gain (58.37 dB) and lower CMRR (88.17 dB), consistent with increased device speed and reduced intrinsic gain.
- The slow–fast (SF) corner results in moderate gain (63.2 dB) and a CMRR of 90.52 dB, reflecting performance degradation due to process mismatch between NMOS and PMOS devices.
- The fast–slow (FS) corner fails to produce valid results, suggesting possible biasing or headroom limitations in the folded cascode structure mosfets are out of saturation.
- Overall, the folded cascode OTA demonstrates significant sensitivity to process variations, with gain and CMRR degrading notably in mismatch and fast-process corners.

– Power Consumption = $16.8\,\mu\text{W}$